

Class 9: Candy Mini Project

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Background

Importing Candy Data, the data comes as a CSV file from 538.

```
candy_file <- read.csv("https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-  
candy = read.csv( "https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power-
```

Q1. How many different candy types are in this dataset?

```
nrow(candy_file)
```

```
[1] 85
```

There are 85 different candy types in this dataset.

Q2. How many fruity candy types are in the dataset?

```
table(candy_file$fruity)
```

```
0 1
47 38
```

There are 38 fruity candy types.

Winpercent

```
candy_file["Twix", ]$winpercent
```

```
[1] 81.64291
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy_file |>
  filter(row.names(candy_file)=="Twix") |>
  select(winpercent)
```

```
      winpercent
Twix    81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is its winpercent value?

My favorite candy is Snickers and the winpercent value is 76.67378.

```
candy_file["Snickers", ]$winpercent
```

```
[1] 76.67378
```

Q4. What is the winpercent value for “Kit Kat”?

The winpercent value for Kit Kat is 76.7686.

```
candy_file["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

The winpercent value for Tootsie Roll Snack Bars is 49.6535.

```
candy_file["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

```
library("skimr")
skim(candy_file)
```

Table 1: Data summary

Name	candy_file
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

The “n_missing” and “complete_rate” columns are on a 0 or 1 scale instead of 0 to 1 values.

Q7. What do you think a zero and one represent for the candy\$chocolate column?

I think a 0 represents if its not a chocolate based candy and 1 represents if it is a chocolate based candy.

Exploratory analysis

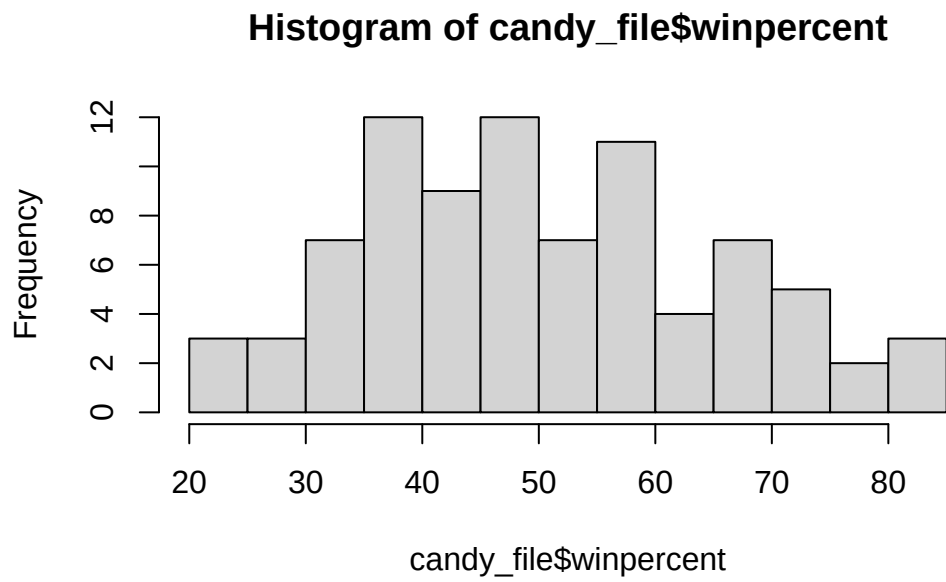
We are going to start by making histograms both with the base R functions and with ggplot.

Q8. Plot a histogram of winpercent values using both base R and ggplot2.

These are below!

Base R histogram:

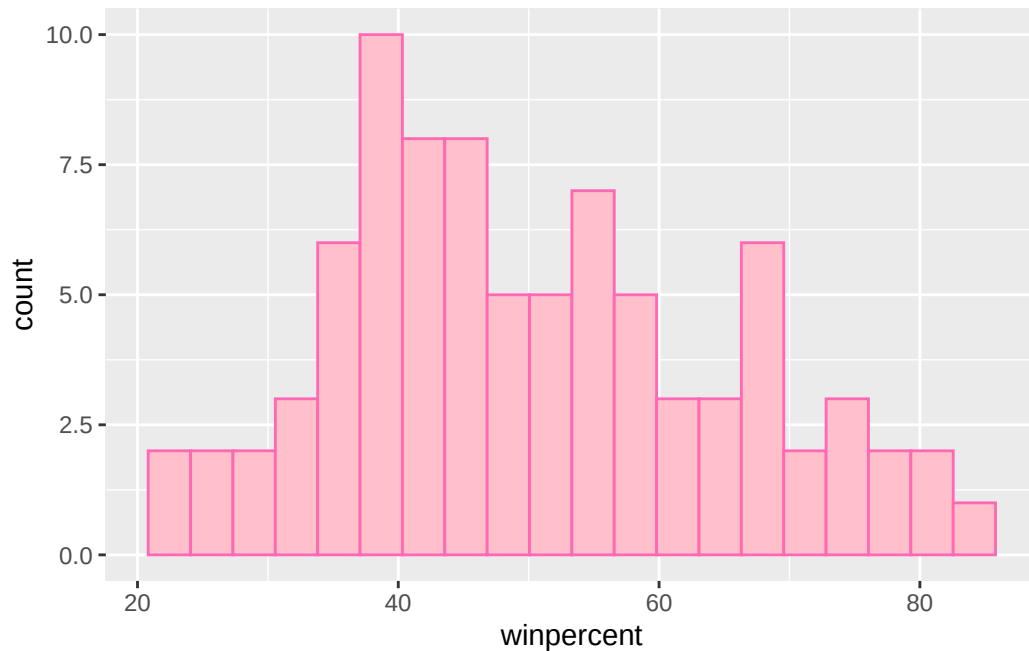
```
hist(candy_file$winpercent, breaks = 20)
```



This is the ggplot one!

```
library(ggplot2)

ggplot(candy_file) +
  aes(x = winpercent) +
  geom_histogram(bins = 20, fill = "pink", col = "hotpink")
```



Q9. Is the distribution of winpercent values symmetrical?

No, the distribution does not appear symmetrical.

Q10. Is the center of the distribution above or below 50%?

No, the mean is above 50% (50.31676% to be exact).

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

On average, chocolate candy is ranked higher than fruit candy.

Steps to solve this: 1. Find all chocolate candy in the dataset. 2. Extract or find their winpercent values. 3. Calculate the mean of these values. 4. Find all fruit candy. 5. Find their winpercent values. 6. Calculate their mean value.

```
choc.candy <- candy_file[ candy_file$chocolate == 1, ]
choc.win <- choc.candy$winpercent
mean(choc.win)
```

```
[1] 60.92153
```

```
fruit.candy <- candy_file[ candy_file$fruity == 1, ]
fruit.win <- fruit.candy$winpercent
mean(fruit.win)
```

```
[1] 44.11974
```

Q12. Is this difference statistically significant?

Yes the difference is statistically significant, the p-value is less than 0.05.

```
t.test(choc.win, fruit.win)
```

Welch Two Sample t-test

```
data: choc.win and fruit.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Overall Candy Ranking

Q13. What are the five least liked candy types in this set?

The five least liked candy types are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

```
head(candy_file[order(candy_file$winpercent),], n=5)
```

	chocolate	fruity	caramel	peanut	almond	nougat		
Nik L Nip	0	1	0		0	0		
Boston Baked Beans	0	0	0		1	0		
Chiclets	0	1	0		0	0		
Super Bubble	0	1	0		0	0		
Jawbusters	0	1	0		0	0		

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip		0	0	0		1		0.197		0.976
Boston Baked Beans		0	0	0		1		0.313		0.511
Chiclets		0	0	0		1		0.046		0.325
Super Bubble		0	0	0		0		0.162		0.116
Jawbusters		0	1	0		1		0.093		0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Q14. What are the top 5 all time favorite candy types out of this set?

The top 5 candy types are Reese's, Reese's minis, Kit Kat, Twix, and Snickers.

```
head(candy_file[order(candy_file$winpercent, decreasing = TRUE), ], n = 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

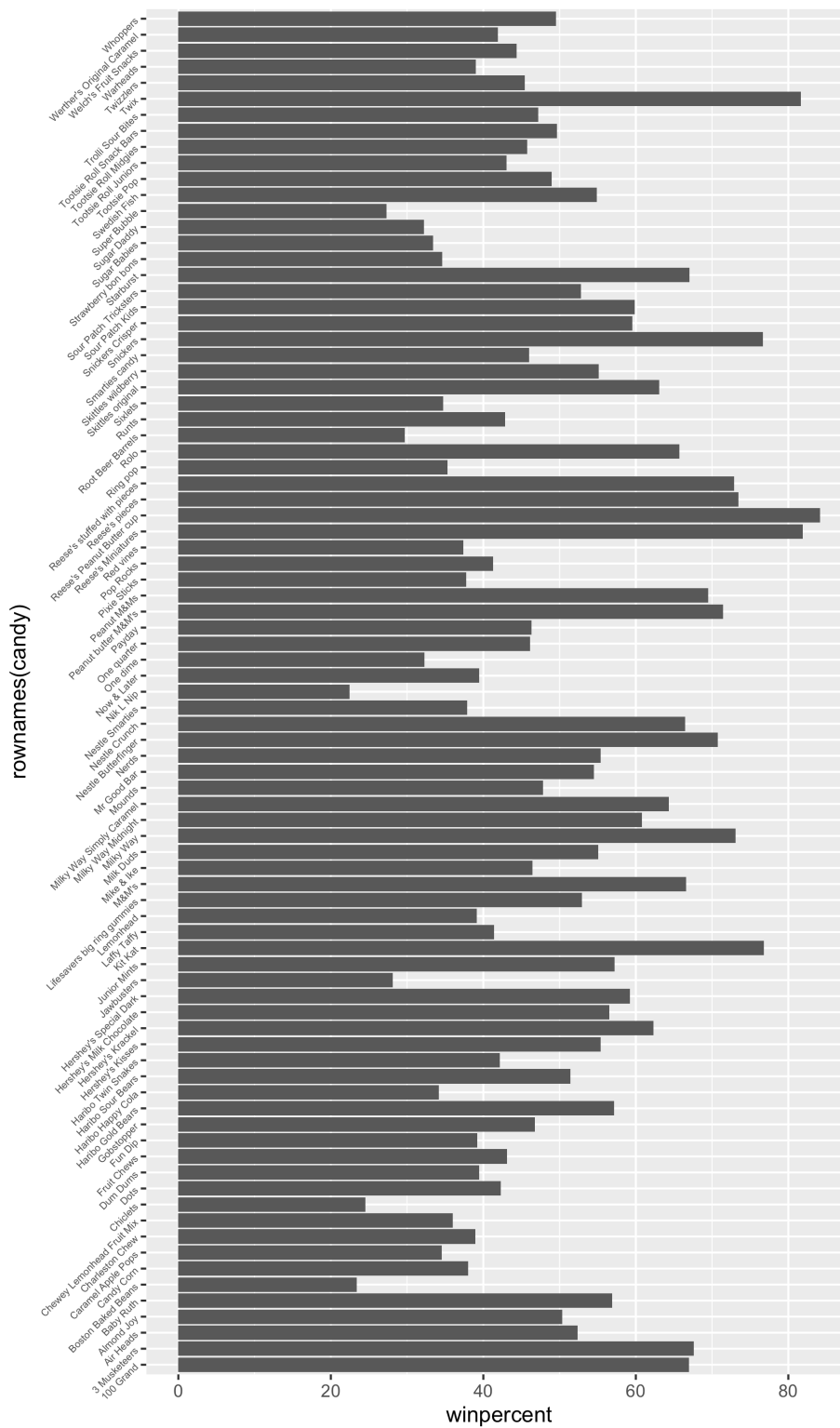
	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Reese's Peanut Butter cup		0	0	0		0		0.720
Reese's Miniatures		0	0	0		0		0.034
Twix		1	0	1		0		0.546
Kit Kat		1	0	1		0		0.313
Snickers		0	0	1		0		0.546

	price	percent	winpercent
Reese's Peanut Butter cup	0.651		84.18029
Reese's Miniatures	0.279		81.86626
Twix	0.906		81.64291
Kit Kat	0.511		76.76860
Snickers	0.651		76.67378

Now we are going to make a bar plot to examine the rankings based on winpercent values.

```
library(ggplot2)

ggplot(candy_file) +
  aes(winpercent, rownames(candy)) +
  geom_col() +
  theme(axis.text.y = element_text(angle = 45, hjust = 1, size = 5))
```

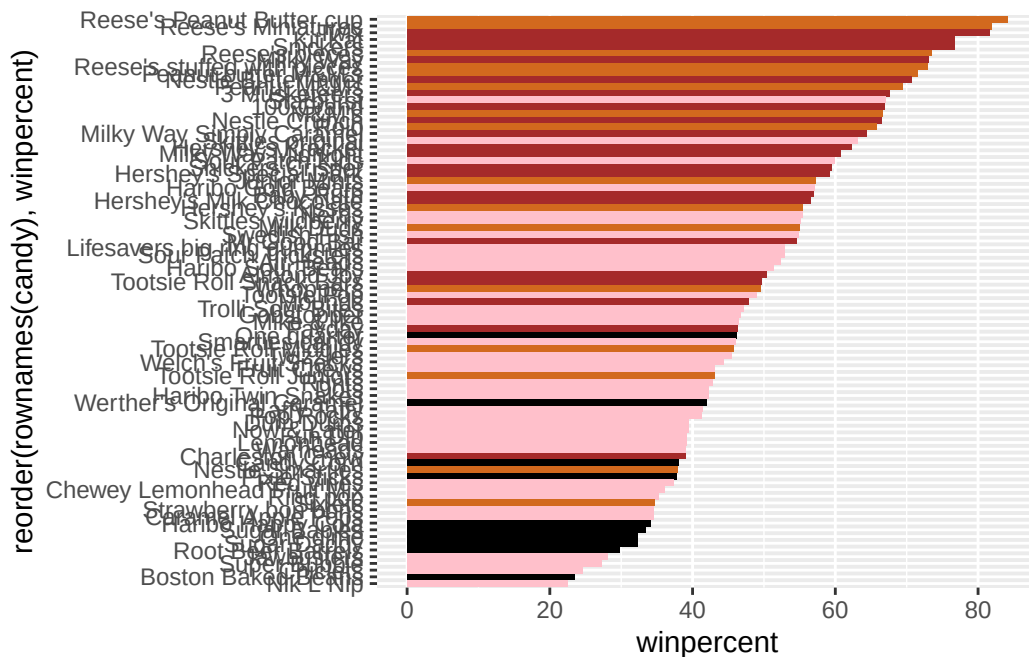



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by `winpercent`?

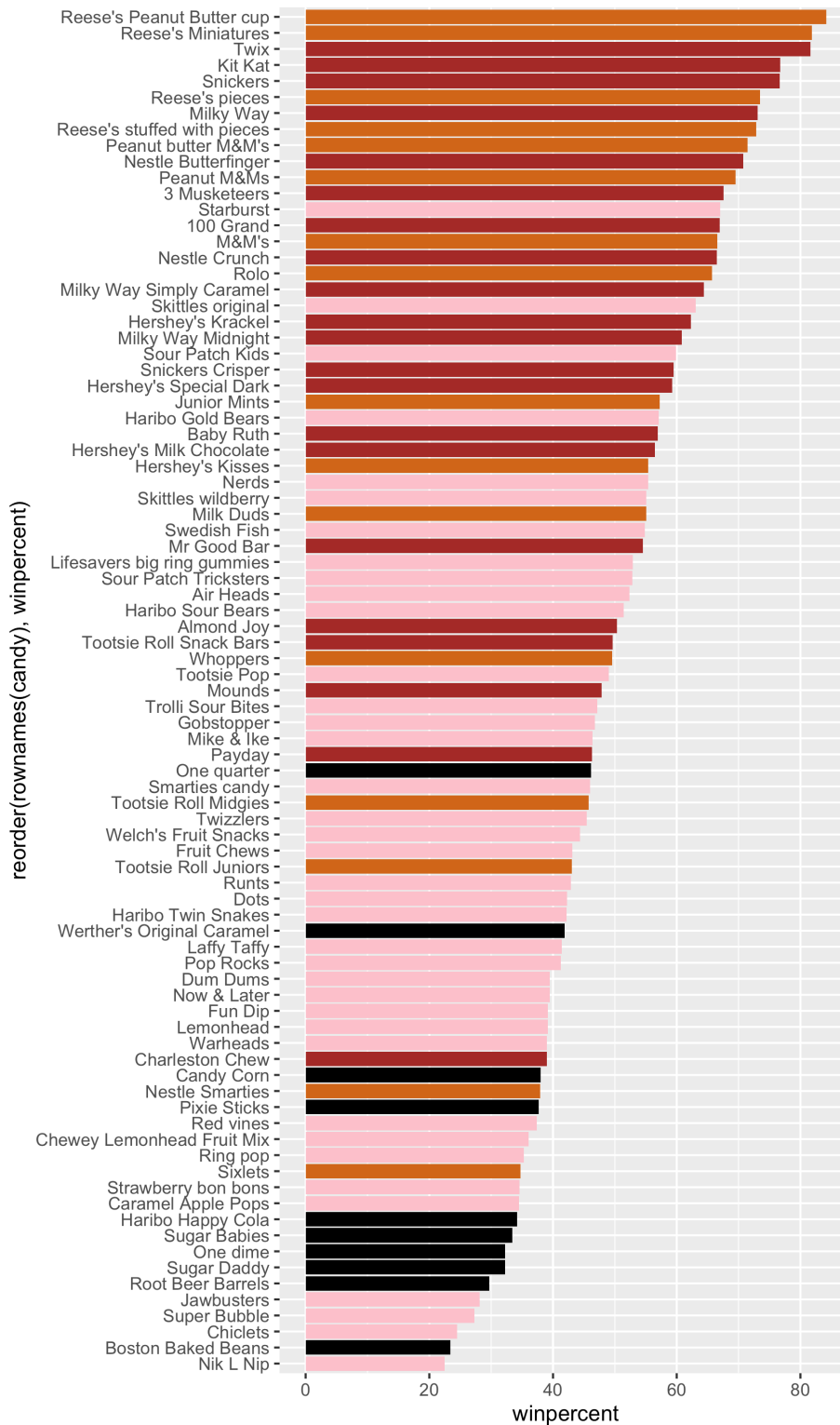
```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"
```

```
library(ggplot2)

ggplot(candy_file) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill=my_cols)
```



```
ggsave("barplot2.png", height = 10, width = 6)
```



Q17. What is the worst ranked chocolate candy?

Sixlets are the worst ranked chocolate candy.

Q18. What is the best ranked fruity candy?

The best ranked fruity candy is Starburst.

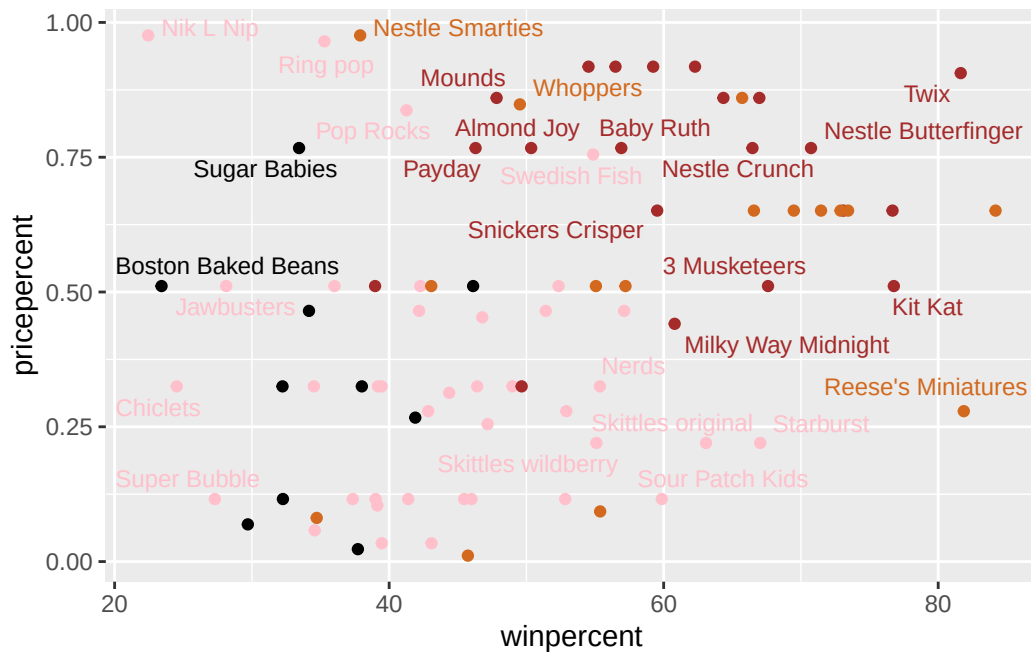
Taking a look at Pricepercent

We can use the **ggrepel** package for better label placement.

```
library(ggrepel)

ggplot(candy_file) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 7)
```

Warning: ggrepel: 57 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

I think it would be Reese's miniatures, since it's far over in winpercent but also not that high for pricepercent.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

The top 5 most expensive candy types are Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, and Hershey's Milk Chocolate. Of these, the least popular is Nik L Nip is the least popular.

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

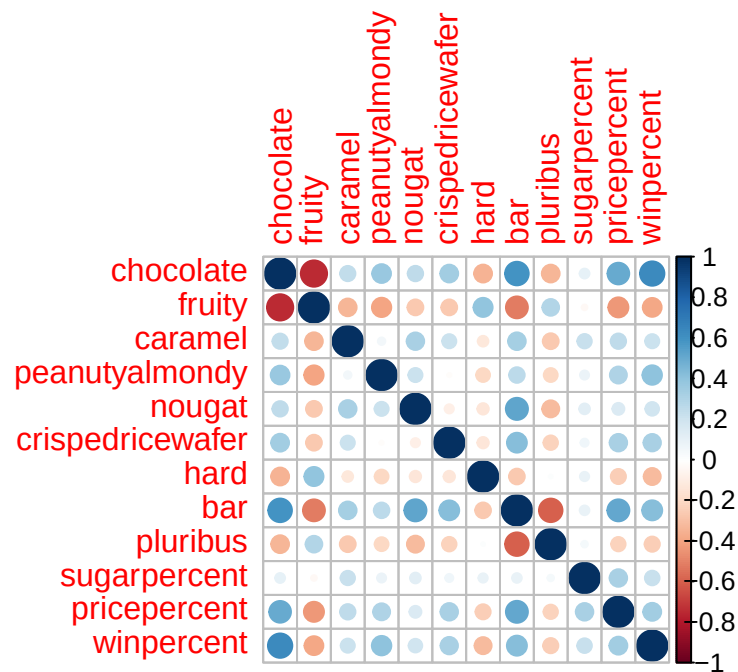
	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

Exploring the Correlation structure

```
library(corrplot)
```

corrplot 0.95 loaded

```
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Chocolate and Fruity are negatively correlated.

```
cor(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
chocolate	1.0000000	-0.74172106	0.24987535	0.37782357	0.25489183
fruity	-0.7417211	1.00000000	-0.33548538	-0.39928014	-0.26936712
caramel	0.2498753	-0.33548538	1.00000000	0.05935614	0.32849280
peanutyalmondy	0.3778236	-0.39928014	0.05935614	1.00000000	0.21311310
nougat	0.2548918	-0.26936712	0.32849280	0.21311310	1.00000000
crispedricewafer	0.3412098	-0.26936712	0.21311310	-0.01764631	-0.08974359
hard	-0.3441769	0.39067750	-0.12235513	-0.20555661	-0.13867505
bar	0.5974211	-0.51506558	0.33396002	0.26041960	0.52297636
pluribus	-0.3396752	0.29972522	-0.26958501	-0.20610932	-0.31033884
sugarpercent	0.1041691	-0.03439296	0.22193335	0.08788927	0.12308135
pricepercent	0.5046754	-0.43096853	0.25432709	0.30915323	0.15319643
winpercent	0.6365167	-0.38093814	0.21341630	0.40619220	0.19937530

	crispedricewafer	hard	bar	pluribus
chocolate	0.34120978	-0.34417691	0.59742114	-0.33967519
fruity	-0.26936712	0.39067750	-0.51506558	0.29972522

caramel	0.21311310	-0.12235513	0.33396002	-0.26958501
peanutyalmondy	-0.01764631	-0.20555661	0.26041960	-0.20610932
nougat	-0.08974359	-0.13867505	0.52297636	-0.31033884
crispedricewafer	1.00000000	-0.13867505	0.42375093	-0.22469338
hard	-0.13867505	1.00000000	-0.26516504	0.01453172
bar	0.42375093	-0.26516504	1.00000000	-0.59340892
pluribus	-0.22469338	0.01453172	-0.59340892	1.00000000
sugarpercent	0.06994969	0.09180975	0.09998516	0.04552282
pricepercent	0.32826539	-0.24436534	0.51840654	-0.22079363
winpercent	0.32467965	-0.31038158	0.42992933	-0.24744787
	sugarpercent	pricepercent	winpercent	
chocolate	0.10416906	0.5046754	0.6365167	
fruity	-0.03439296	-0.4309685	-0.3809381	
caramel	0.22193335	0.2543271	0.2134163	
peanutyalmondy	0.08788927	0.3091532	0.4061922	
nougat	0.12308135	0.1531964	0.1993753	
crispedricewafer	0.06994969	0.3282654	0.3246797	
hard	0.09180975	-0.2443653	-0.3103816	
bar	0.09998516	0.5184065	0.4299293	
pluribus	0.04552282	-0.2207936	-0.2474479	
sugarpercent	1.00000000	0.3297064	0.2291507	
pricepercent	0.32970639	1.0000000	0.3453254	
winpercent	0.22915066	0.3453254	1.0000000	

Q23. Similarly, what two variables are most positively correlated?

Variables that are the same are the most positively correlated, so Chocolate and chocolate and fruity with fruity.

Principal Component Analysis

```
pca <- prcomp(candy, scale = T)
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369
	PC8	PC9	PC10	PC11	PC12		
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760		

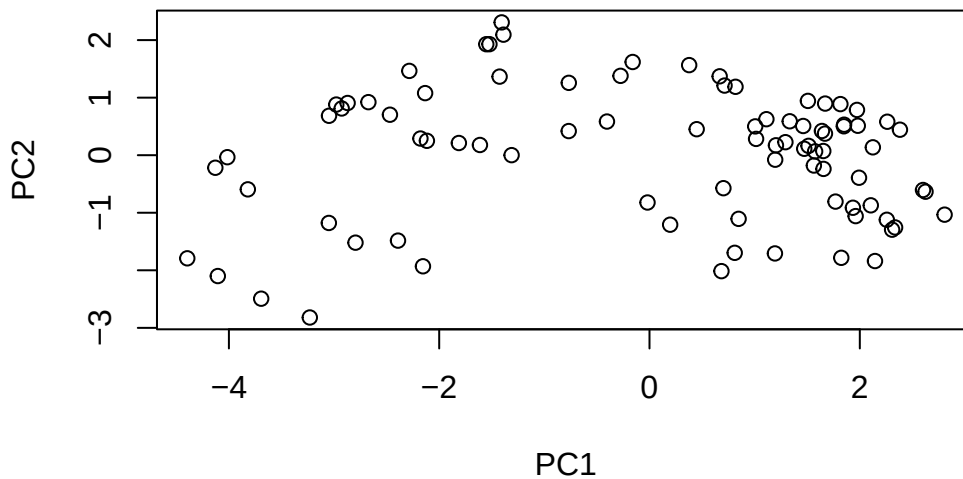
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
pca$rotation[,1]
```

chocolate	fruity	caramel	peanutyalmondy
-0.4019466	0.3683883	-0.2299709	-0.2407155
nougat	crispedricewafer	hard	bar
-0.2268102	-0.2215182	0.2111587	-0.3947433
pluribus	sugarpercent	pricepercent	winpercent
0.2600041	-0.1083088	-0.3207361	-0.3298035

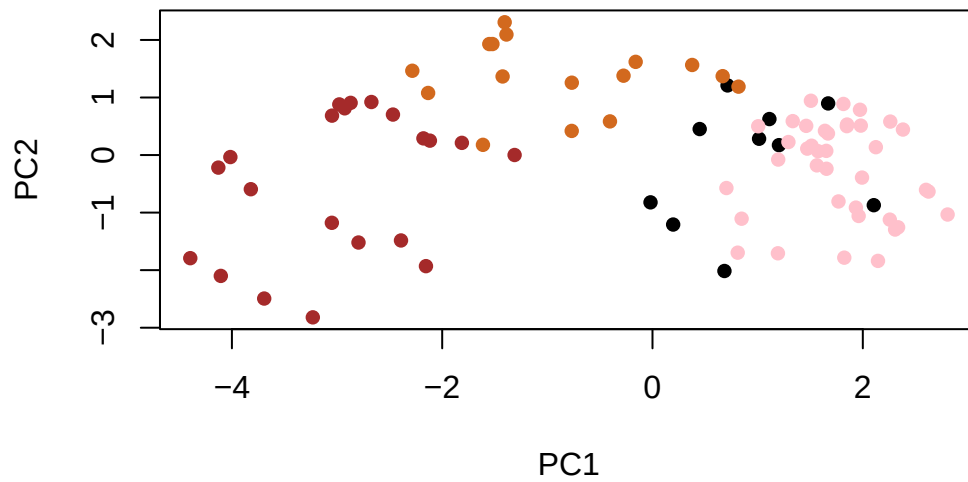
Now we can plot!

```
plot(pca$x[, 1:2])
```



We can change the plotting character and add some color!

```
plot(pca$x[,1:2], col=my_cols, pch=16)
```

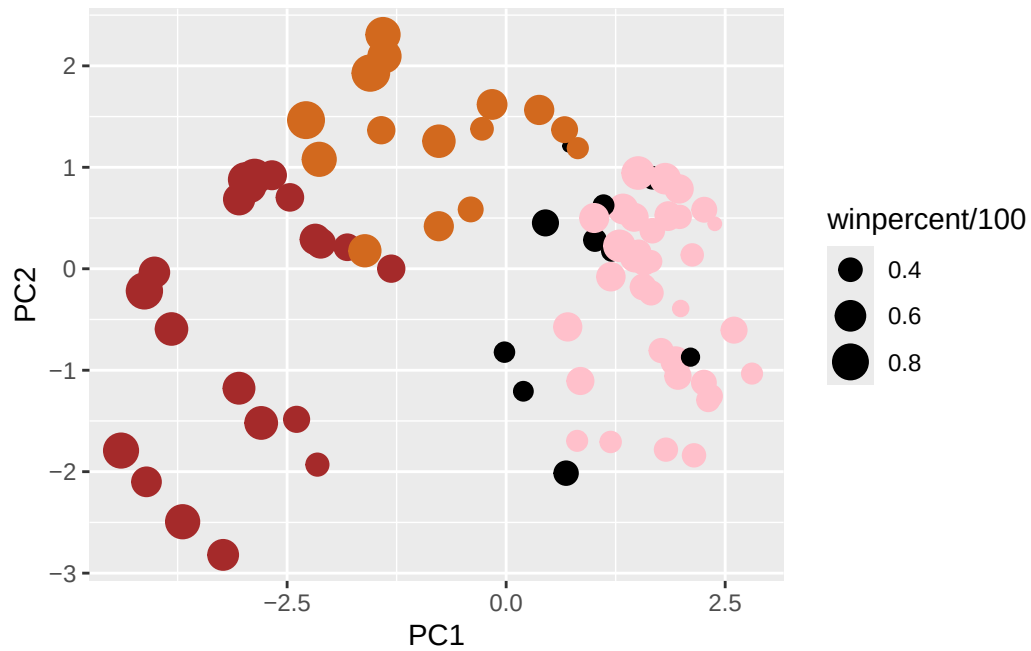


Main result figure:

```
my_data <- cbind(candy, pca$x[,1:3])

p <- ggplot(my_data) +
  aes(PC1, PC2,
      size=winpercent/100,
      text=rownames(my_data),
      label=rownames(my_data)) +
  geom_point(col = my_cols)

p
```



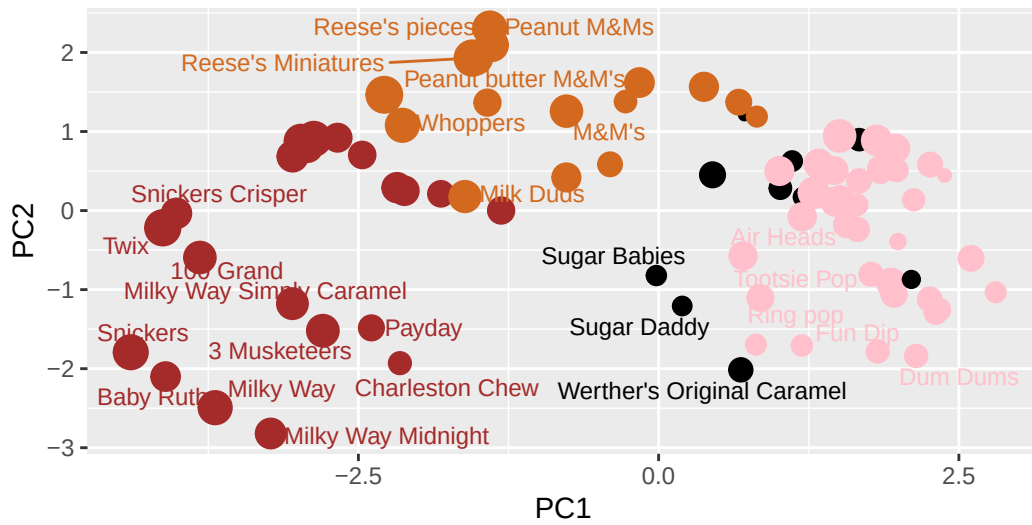
```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
        subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
        caption="Data from 538")
```

Warning: ggrepel: 59 unlabeled data points (too many overlaps). Consider increasing max.overlaps

Halloween Candy PCA Space

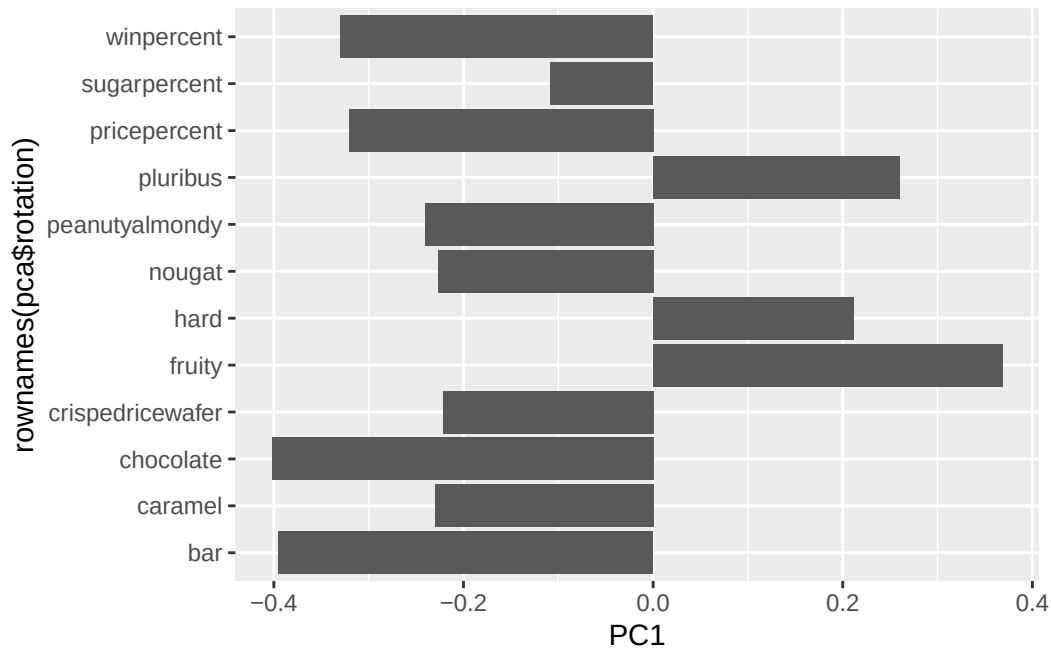
Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

```
#library(plotly)
#ggplotly(p)
```

```
ggplot(pca$rotation) +
  aes(x = PC1, y = rownames(pca$rotation)) +
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

Pluribus, hard, and fruity are the original variables picked up strongly by PC1. This makes sense to me, we previously saw this with the correlation structure.

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

Chocolate candy is preferred, we saw this in our `t.test`, if it's pluribus it's also preferred as seen in the PCA analysis.